

Humphrey Curtis

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London, UK · Open to relocation

Research Engineer & Applied Scientist specialising in AI for accessibility and socially disadvantaged populations. PhD in Computer Science from King's College London, awarded the 2026 ACM SIGCHI Outstanding Dissertation Award; formerly a Research Intern at Microsoft Research. Fifteen peer-reviewed papers including two ACM Best Paper Awards, a Scientific Reports Nature article and a Routledge book chapter. Research spans wearable augmentative and alternative communication (AAC) tools co-designed with 200+ stroke survivors — smartwatch, smartbadge, iOS, and HoloLens — through to production GenAI systems for welfare navigation: schema-driven LLM pipelines and adversarial guardrails, adopted nationally by the UK Stroke Association & DWP.

EXPERIENCE

King's College London, Research Associate London, UK | 11/2025–Present

- Designed the LLM prompting layer and schema-driven agentic architecture for *Advokit* (advokit.uk) — a GenAI welfare-navigation toolkit where each page regenerates from a JSON schema co-editable by users and the LLM. Engineered structured reasoning chains, adversarial stress-tests and safety guardrails for a production system serving vulnerable users.
- Built evaluation infrastructure (800 visitors, 2,849 page views) surfacing LLM failure modes, driving redesign toward agentic interactions. Translated R&D in stakeholder workshops with the Stroke Association and DWP; adopted for UK-wide rollout.

Microsoft Research, Research Intern, Ability Team Seattle, USA | 06/2024–09/2024

- Designed and evaluated LLM-powered agentic meeting-assistant prototypes for Microsoft Teams, supporting chronically ill users pre-, during and post-call. First-authored CHI '25 paper; co-authored MSR tech report on AAC & agentic AI.
- Led the largest study of its kind (N=55) on video-calling experiences of chronically ill people; mixed-methods synthesis produced 8 design recommendations, presented to product and engineering teams.

King's College London, PhD Researcher London, UK | 2021–2026

- Designed, built and deployed AI-driven communication tools across iOS, watchOS, e-ink (Badger2040) and mixed reality (HoloLens); co-designed with 200+ stroke survivors, clinicians and SLTs over four years (open-sourcing on GitHub).
- Developed evaluation frameworks for communicative confidence, felt understanding and social presence. Research contributed to a £2M UKRI Future Leaders Fellowship.

RESEARCH

Google Scholar | ORCID

16 peer-reviewed papers (11 first-author) at ACM CHI & ASSETS; *Scientific Reports* (Nature); Routledge book chapter (Taylor & Francis). Highlights: **Best Paper**, CHI 2023 (top 1% of ~3,500); **Best Student Paper**, ASSETS 2023. **Best Paper Nomination**, ASSETS 2024.

Invited talks: DWP (2026); Stroke Association (2026); Google HQ (2025); Univ. of Oklahoma (2025); Microsoft Research (2024); Transport for London (2023). **Press:** *Electronic Specifier Magazine*; *Enabled in Academia* (2023).

TECHNICAL SKILLS

AI & LLM: Prompt engineering (adversarial stress-testing, guardrail design, reasoning chains), schema-driven agentic pipelines, evaluation & monitoring, safety design for vulnerable users, on-device ML

Research Methods: Participatory design, mixed-methods, usability testing, diary studies, A/B testing, survey design, conversation analysis

Programming & Tools: Python, TypeScript/JavaScript (React, Next.js), Swift, HTML/CSS, SQL, C; MongoDB, Firebase, REST APIs, Docker, Git; Figma, WCAG, D3.js

EDUCATION

King's College London, PhD Computer Science (Awarded 2026) 2021–2026

Thesis: Wearable & Discreet Augmentative & Alternative Communication.

King's College London, MSc Advanced Computer Science, Distinction 2020–2021

University of Bristol, MSc Computer Science, Merit 2019–2020

University of Exeter, BA Politics, Philosophy & Economics, First Class 2015–2019

SELECTED HONOURS

2026 ACM SIGCHI Outstanding Dissertation Award — top doctoral thesis in HCI globally

2026 King's Education Award Nomination, Colleagues' Choice

2025 Ken Rothman Award, Google HQ — Health Research From Home

2024 Best Paper Nomination, ACM ASSETS (top 5 of ~220 submissions)

2024 NMES PhD Research Competition Award (faculty-wide, science, maths & engineering)

2023 Best Paper Award, ACM CHI (top 1% of ~3,500 submissions)

2023 Best Student Paper, ACM ASSETS (top 2 of ~180 submissions)

2021 KCL PhD Studentship — departmental scholarship covering fees and living costs

2021 MSc Advanced Computing Academic Prize — top student across entire cohort

SERVICE

ACM ASSETS '26: Accessibility Co-Chair; AI Advisory Committee Member Porto, Portugal

ACM ASSETS '24: Student Volunteer Chair; Research Mentor St John's, Canada

BCS HCI: Associate Chair (2025) · **Cockney CHAI:** Web Chair (2025)

Peer Review: CHI, UIST, ASSETS, TACCESS, TOG, AAC Journal & 4 further venues (2023–Present)

Volunteer: Aphasia Re-Connect, Speech & Language Therapy Centre, London (2021–Present)

Longer CV →